

IT694 Computer Networks (Winter'2026)

Instructor: P.S. Kalyan Sasidhar, 2109, Faculty Block-2

Course Placement: MSc IT, Semester-II

Course format: 3 Lecture hours and 2 Lab hours.

Course Content:

This course will cover the fundamental principles of wired computer networks with focus on layered architecture, protocols, implementation and issues specific to the Internet. The objective is to provide an understanding of how the Internet works which include how data flows from a source to a destination. Focus will also be laid on certain topics such as network design and automation, network management, administration and debugging which are ideally required for networking IT professionals.

The associated laboratory component is designed to expose students to the basic LAN hardware and configuration, and network simulation tools for the analysis of traffic and network protocols. Cisco laboratory exercises will be strongly referred to provide an industry level exposure to real life networking scenarios based labs.

Textbook:

1. James F. Kurose and Keith W. Ross. 2016. *Computer Networking: A Top-Down Approach (7th ed.)*. Pearson.

References:

1. Douglas E. Comer. 2013. *Internetworking with TCP/IP (6th ed.)*. Addison-Wesley Professional.
2. Andrew S. Tanenbaum and David J. Wetherall. 2010. *Computer Networks (5th ed.)* Prentice Hall Press, Upper Saddle River, NJ, USA.
3. Larry L. Peterson and Bruce S. Davie. 2011. *Computer Networks, Fifth Edition: A Systems Approach (5th ed.)*. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA

Evaluation Plan:

In-Sem I	15%
In-Sem II	20%
End sem	35%
Labs	25%
Quizzes	5%

Course outcomes:

At the end of the course, students are expected to:

1. Understand the TCP/IP Internet reference Model
2. Learn commonly used network protocols and their design
3. Be able to design and develop network applications
4. Be exposed to traffic engineering and multimedia networking concepts
5. Measure performance of protocols using analytical methods and simulation tools

Program outcomes

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X			X	X				X			

Tentative Lecture Outline:

Sl No	Description		Lectures
1	Overview		6
	1.1	Overview	
	1.2	Internet-Birds' Eye View, History	
	1.3	Internet-Layered Architecture:OSI and TCP/IP architectures	
	1.4	Packet Switching, Best Effort Services	
2	Network Applications		6
	2.1	Client-Server Applications	
	2.2	Chat Application Design, Socket Programming	
	2.3	SFTP File Transfer Protocol	
	2.4	Domain Name Service, Mail, SMTP	
	2.5	Peer to Peer Search, Distributed Hash	
3	End to End Issues		6
	3.1	Transport Layer Basics	
	3.2	Reliability	
	3.3	Connectionless and Connection Oriented Transport	
	3.4	TCP and UDP protocols	
	3.5	Congestion Management	
	3.6	TCP Performance Measure	
4	Routing and Congestion		8
	4.1	Introduction	
	4.2	Scheduling, Best Effort Service	
	4.3	Scheduling for Guaranteed Service, Switching	
	4.4	Packet Switching, Batched Banyan Switch	
	4.5	Routing - Introduction, Multicast, Broadcast	
	4.6	Addressing, CIDR, IP Protocol IPv4, IPv6	
	4.7	Hierarchical Routing, BGP, Mobile Routing	
	4.8	Control and Data Path, Open Flow	

	4.9	Software Defined Networking	
5	Link Layer Technologies		6
	5.1	Introduction	
	5.2	Media Access Protocols, ALOHA	
	5.3	IEEE 802.3 Ethernet Protocol, MACA	
	5.4	Switched LAN, Virtual LANs	

Sl No	Description		Lectures
6	Network Design and Automation		4
	6.1	Network Automation and Programmability Tools and Techniques (Python, Ansible, Puppet and Chef)	
	6.2	Application Programming Interface (RESTful API request and response)	
	6.3	Green and Sustainable Network Infrastructure Design (such as advanced cooling techniques and power management)	
7	Content Distribution Networks		3
	7.1	Architecture and protocols	
	7.2	Video Streaming, DASH	
8	Network Management		3
	8.1	Network Debugging	
	8.2	Revision	